MA563 HW3

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Question 1

Use Taylor series to derive a four-point formula for $f'(0)$ using the nodes $x = 0, -h, -2h, -3h$. Determine the degree of accuracy and the order of accuracy.

We seek a linear combination of function values such that:

$$f'(0) \approx Af(0) + Bf(-h) + Cf(-2h) + Df(-3h)$$

We expand each term around $x = 0$ using Taylor's Theorem:

$$f(-h) = f(0) - hf'(0) + \frac{h^2}{2}f''(0) - \frac{h^3}{6}f'''(0) + \frac{h^4}{24}f^{(4)}(0) + O(h^5)$$

$$f(-2h) = f(0) - 2hf'(0) + \frac{4h^2}{2}f''(0) - \frac{8h^3}{6}f'''(0) + \frac{16h^4}{24}f^{(4)}(0) + O(h^5)$$

$$f(-3h) = f(0) - 3hf'(0) + \frac{9h^2}{2}f''(0) - \frac{27h^3}{6}f'''(0) + \frac{81h^4}{24}f^{(4)}(0) + O(h^5)$$

Substituting these expansions into our approximation formula and grouping by derivatives of f at 0:

- Terms of $f(0)$: $A + B + C + D = 0$
- Terms of $f'(0)$: $-hB - 2hC - 3hD = 1$
- Terms of $f''(0)$: $\frac{h^2}{2}B + \frac{4h^2}{2}C + \frac{9h^2}{2}D = 0$
- Terms of $f'''(0)$: $-\frac{h^3}{6}B - \frac{8h^3}{6}C - \frac{27h^3}{6}D = 0$

Dividing out the h factors, we obtain the system:

1. $A + B + C + D = 0$
2. $B + 2C + 3D = -\frac{1}{h}$
3. $B + 4C + 9D = 0$
4. $B + 8C + 27D = 0$

Solving the system of equations

- From (3) and (4): $4C + 18D = 0 \implies C = -4.5D$

- Substituting into (2) and (3): $\mathbf{D} = -\frac{\mathbf{1}}{\mathbf{3h}}$, $\mathbf{C} = \frac{\mathbf{3}}{\mathbf{2h}}$, $\mathbf{B} = -\frac{\mathbf{3}}{\mathbf{h}}$, $\mathbf{A} = \frac{\mathbf{11}}{\mathbf{6h}}$

Substituting the coefficients back into the original expression:

$$f'(0) = \frac{11f(0) - 18f(-h) + 9f(-2h) - 2f(-3h)}{6h} + E(f)$$

To find the error term $E(f)$, look at the first non-vanishing term in the Taylor expansion, which is the $f^{(4)}(0)$ term:

$$E(f) = \left[B\frac{h^4}{24} + C\frac{16h^4}{24} + D\frac{81h^4}{24}\right] f^{(4)}(\eta)$$

Substituting our values for B, C, and D:

$$E(f) = \frac{h^4}{24}\left[-\frac{3}{h} + \frac{24}{h} - \frac{27}{h}\right] f^{(4)}(\eta) = -\frac{6h^3}{24} f^{(4)}(\eta)$$

$$\mathbf{E(f)} = -\frac{\mathbf{1}}{\mathbf{4}}\mathbf{h^3 f^{(4)}}(\eta)$$

- Degree of Accuracy: **3** (The formula is exact for polynomials $P(x)$ where $\deg(P) \leq 3$).
- Order of Accuracy: $O(h^3)$ (Third-order accurate).

Question 2 Richardson extrapolation

We are given a three-point forward difference formula $D(h)$ with an error expansion where the terms do not have a gap in powers (unlike central differences):

$$f'(x) = D(h) + c_2h^2 + c_3h^3 + c_4h^4 + \dots$$

The goal is to eliminate the leading error terms sequentially using Richardson Extrapolation.

(a) Derivation of $D_1(h)$

To eliminate the h^2 term, we compare $D(h)$ and $D(h/2)$:

1. $f'(x) = D(h) + c_2h^2 + c_3h^3 + \dots$
2. $f'(x) = D(h/2) + c_2(\frac{h}{2})^2 + c_3(\frac{h}{2})^3 + \dots = D(h/2) + \frac{c_2}{4}h^2 + \frac{c_3}{8}h^3 + \dots$

We seek a combination $D_1(h) = BD(h/2) + AD(h)$ such that the c_2 terms cancel:

- $A + B = 1$
- $Ac_2h^2 + B\frac{c_2}{4}h^2 = 0 \implies A + \frac{1}{4}B = 0$

$B = \frac{4}{3}$ and $A = -\frac{1}{3}$.

Thus:

$$\mathbf{D_1(h)} = \frac{\mathbf{4D(h/2) - D(h)}}{\mathbf{3}}$$

New Error Term:

The h^3 term becomes: $\left(\frac{4}{3} \cdot \frac{c_3}{8} - \frac{1}{3}c_3\right) h^3 = -\frac{1}{6}c_3h^3$. So, $D_1(h) = f'(x) + O(h^3)$.

(b) General Recurrence for $D_k(h)$

Assume that after k extrapolations, the approximation $D_k(h)$ has an error order of $O(h^{k+2})$:

$$f'(x) = D_k(h) + c_{k+2}h^{k+2} + O(h^{k+3})$$

Following the same logic to eliminate the h^{k+2} term, we solve for weights A_k and B_k:

- $A_k + B_k = 1$
- $A_k(1) + B_k(\frac{1}{2})^{k+2} = 0$

This gives:

$$B_k = \frac{2^{k+2}}{2^{k+2} - 1}, \quad A_k = -\frac{1}{2^{k+2} - 1}$$

For $k = 0, 1, 2, \ldots$ with $D_0(h) = D(h)$:

$$\mathbf{D_{k+1}(h) = \frac{2^{k+2}D_k(h/2) - D_k(h)}{2^{k+2} - 1}}$$

Each subsequent $D_k(h)$ is an approximation of order $O(h^{k+2})$.

Question 3

(a) Symmetry of the Nodal Polynomial

We consider the closed Newton-Cotes formula with nodes $x_j = a + jh$ for $j = 0, \ldots, n$ and $h = (b - a)/n$. For an even n, we aim to show that the integral of the nodal polynomial $\omega_n(x) = \prod_{j=0}^{n}(x - x_j)$ over $[a, b]$ is zero.

Proof: Let $c = (a + b)/2$ be the midpoint of the interval. Since n is even, c is a node ($x_{n/2}$). Define the relative distances $d_j = x_j - c$. By symmetry, $d_{n-j} = -d_j$.

Substituting $x = c + u$:

$$\omega_n(c + u) = \prod_{j=0}^{n}(c + u - x_j) = \prod_{j=0}^{n}(u - d_j)$$

Now consider $\omega_n(c - u)$:

$$\omega_n(c - u) = \prod_{j=0}^{n}(c - u - x_j) = \prod_{j=0}^{n}(-u - d_j) = (-1)^{n+1}\prod_{j=0}^{n}(u + d_j)$$

Since the set $\{d_j\}$ is symmetric about zero, $\prod(u + d_j) = \prod(u - d_j)$. For even n, the exponent $n + 1$ is odd, making $(-1)^{n+1} = -1$. Thus:

$$\omega_n(c - u) = -\omega_n(c + u)$$

This proves ω_n is an odd function about c. Integrating over the symmetric interval $[a, b]$:

$$\int_a^b \omega_n(x)\,dx = \int_{-(b-a)/2}^{(b-a)/2} \omega_n(c + u)\,du = 0$$

(b) Derivation of Simpson's Rule

We approximate $\int_0^1 f(s)\,ds$ using nodes $s_0 = 0, s_1 = \frac{1}{2}, s_2 = 1$.

The Lagrange basis polynomials are:

- $L_0(s) = \frac{(s-1/2)(s-1)}{(0-1/2)(0-1)} = 2(s-1/2)(s-1)$
- $L_1(s) = \frac{(s-0)(s-1)}{(1/2-0)(1/2-1)} = -4s(s-1)$
- $L_2(s) = \frac{(s-0)(s-1/2)}{(1-0)(1-1/2)} = 2s(s-1/2)$

Integrating these weights over $[0,1]$:

- $c_0 = \int_0^1 L_0(s)ds = \frac{1}{6}$
- $c_1 = \int_0^1 L_1(s)ds = \frac{4}{6} = \frac{2}{3}$
- $c_2 = \int_0^1 L_2(s)ds = \frac{1}{6}$

Simpson's Rule: $\int_0^1 f(s)\,ds \approx \frac{1}{6}f(0) + \frac{2}{3}f(1/2) + \frac{1}{6}f(1)$.

Degree of Accuracy:

We test the rule for polynomials $f(s) = s^k$:

- $k = 0$**:** Exact $= 1$, Approx $= 1/6 + 4/6 + 1/6 = 1$ (Exact)
- $k = 1$**:** Exact $= 1/2$, Approx $= 2/3(1/2) + 1/6(1) = 1/2$ (Exact)
- $k = 2$**:** Exact $= 1/3$, Approx $= 2/3(1/4) + 1/6(1) = 1/3$ (Exact)
- $k = 3$**:** Exact $= 1/4$, Approx $= 2/3(1/8) + 1/6(1) = 1/4$ (Exact)

The degree of accuracy is **3**.

(c) Composite Midpoint Rule

Given data at midpoints $x_{i+1/2}$ for $i = 0, \ldots, n-1$ on subintervals of width h:

The composite rule is the sum of local midpoint approximations:

$$\int_a^b f(x)\,dx \approx h \sum_{i=0}^{n-1} f(x_{i+1/2})$$

The local error on $[x_i, x_{i+1}]$ is $\frac{h^3}{24}f''(\xi_i)$. Summing these:

$$E_{total} = \sum_{i=0}^{n-1} \frac{h^3}{24} f''(\xi_i) = \frac{h^3}{24} \cdot n \cdot \left[\frac{1}{n}\sum_{i=0}^{n-1} f''(\xi_i)\right]$$

By the Intermediate Value Theorem, there exists $\eta \in (a, b)$ such that $f''(\eta) = \frac{1}{n}\sum f''(\xi_i)$. Substituting $nh = (b - a)$:

$$E_{total} = \frac{(b-a)h^2}{24} f''(\eta)$$

Bound:

$$|E| \leq \frac{(b-a)h^2}{24} \max_{x\in[a,b]} |f''(x)|$$

Question 4

We estimate the following integrals using the composite trapezoidal rule over the interval $[0, 2\pi]$:

$I_1 = \int_0^{2\pi} \sin^4 x \, dx = \frac{3\pi}{4}$

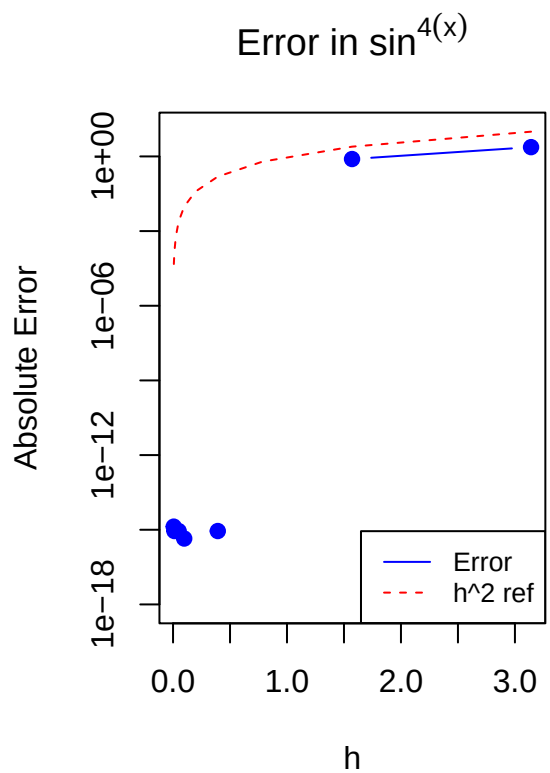
$I_2 = \int_0^{2\pi} \frac{1}{1+16\sin^2 x} \, dx = \frac{2\pi}{\sqrt{17}}$

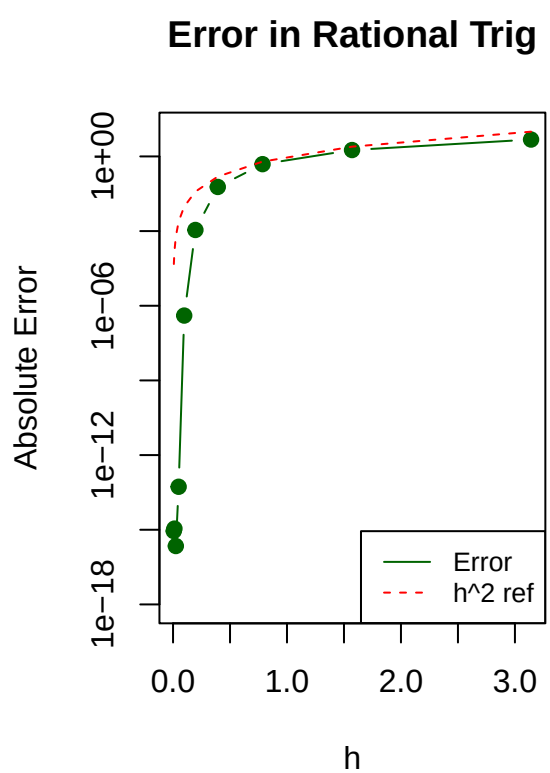
For periodic functions integrated over a full period, the trapezoidal rule simplifies to:

$$T_M = h \sum_{j=0}^{M-1} f(jh), \quad h = \frac{2\pi}{M}$$

```r
n_vals <- 1:10
exact_1 <- 3 * pi / 4
exact_2 <- 2 * pi / sqrt(17)

f1 <- function(x) sin(x)^4
f2 <- function(x) 1 / (1 + 16 * sin(x)^2)

calc_trap_error <- function(f, exact, n) {
  N <- 2^n
  h <- 2 * pi / N
  # Use (0, N-1) to get exactly N unique periodic points
  x <- (0:(N-1)) * h
  approx <- h * sum(f(x))
  return(abs(approx - exact))
}

# Generate Data
h_vals <- (2 * pi) / (2^n_vals)
errors_1 <- sapply(n_vals, function(n) calc_trap_error(f1, exact_1, n))
errors_2 <- sapply(n_vals, function(n) calc_trap_error(f2, exact_2, n))

par(mfrow=c(1, 2), mar=c(5, 4, 4, 2) + 0.1)

# Case (i) Plot
plot(h_vals, errors_1, log="y", type="b", pch=19, col="blue",
     main=expression(paste("Error in ", sin^4(x))),
     xlab="h", ylab="Absolute Error", ylim=c(1e-18, 10))
lines(h_vals, h_vals^2, lty=2, col="red") # Reference slope
legend("bottomright", legend=c("Error", "h^2 ref"), col=c("blue", "red"), lty=1:2, cex=0.8)
```

```r
# Case (ii) Plot
plot(h_vals, errors_2, log="y", type="b", pch=19, col="darkgreen",
     main="Error in Rational Trig",
     xlab="h", ylab="Absolute Error", ylim=c(1e-18, 10))
lines(h_vals, h_vals^2, lty=2, col="red") # Reference slope
legend("bottomright", legend=c("Error", "h^2 ref"), col=c("darkgreen", "red"), lty=1:2, cex=0.8)
```





Observations and Convergence Rates

Case (i): $\sin^4 x$

As seen in the plot, the error follows the $O(h^2)$ reference initially but drops abruptly to machine precision ($\sim 10^{-16}$) at $h \approx 0.785$ ($M = 8$). This is because $\sin^4 x$ is a trigonometric polynomial of degree 4. The trapezoidal rule is exact for trigonometric polynomials when the number of points M exceeds the degree of the polynomial. This demonstrates spectral accuracy.

Case (ii): $(1 + 16\sin^2 x)^{-1}$

The error curve (green) lies consistently below the h^2 reference curve and exhibits a slope that increases as h decreases. The convergenece is much faster than the second order. This might due to the fact that the denominator is positive everywhere. For such functions, the trapozoidal rule exhibits exponential convergence. ($E(h) \sim e^{-c/h}$).

Overall, the composite trapezoidal rule performs significantly better than the standard $O(h^2)$ bound for these periodic cases. While Case (i) achieves exactness once the frequency is resolved, Case (ii) demonstrates the efficiency of spectral methods for smooth, analytic periodic integrands.